

MATH-SHEPHERD: VERIFY AND REINFORCE LLMs STEP-BY-STEP WITHOUT HUMAN ANNOTATIONS

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ABSTRACT

In this paper, we present an innovative process-oriented math process reward model called **MATH-SHEPHERD**, which assigns a reward score to each step of math problem solutions. The training of MATH-SHEPHERD is achieved using automatically constructed process-wise supervision data, breaking the bottleneck of heavy reliance on manual annotation in existing work. We explore the effectiveness of MATH-SHEPHERD in two scenarios: 1) *Verification*: MATH-SHEPHERD is utilized for reranking multiple outputs generated by Large Language Models (LLMs); 2) *Reinforcement Learning*: MATH-SHEPHERD is employed to reinforce LLMs with step-by-step Proximal Policy Optimization (PPO). With MATH-SHEPHERD, a series of open-source LLMs demonstrates exceptional performance. For instance, the step-by-step PPO with MATH-SHEPHERD significantly improves the accuracy of Mistral-7B (77.9%→84.1% on GSM8K and 28.6%→33.0% on MATH). The accuracy can be further enhanced to 89.1% and 43.5% on GSM8K and MATH with the verification of MATH-SHEPHERD, respectively. We believe that automatic process supervision holds significant potential for the future evolution of LLMs.

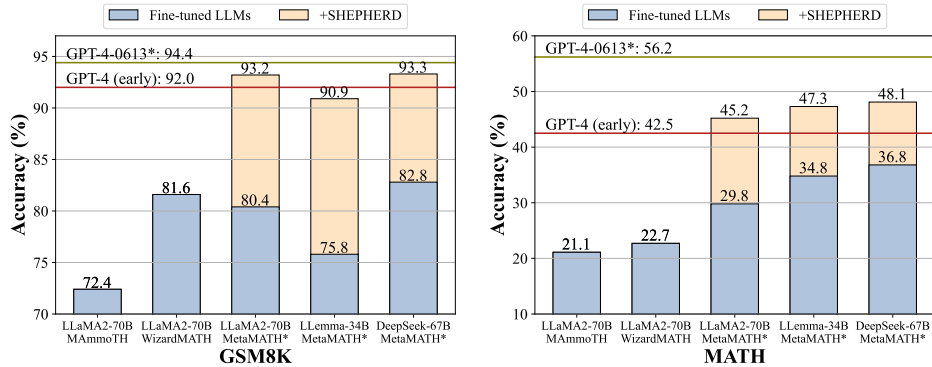


Figure 1: We evaluate the performance of various LLMs with MATH-SHEPHERD on the GSM8K and MATH datasets. All base models are finetuned with the MetaMath dataset (Yu et al., 2023b). The +SHEPHERD results are obtained by selecting the best one from 256 candidates using MATH-SHEPHERD. We observe that MATH-SHEPHERD is compatible with different LLMs. The results of GPT-4 (early) are from Bubeck et al. (2023).

[†]Contribution during internship at DeepSeek-AI.

1 INTRODUCTION

Large language models (LLMs) have demonstrated remarkable capabilities across various tasks (Park et al., 2023; Kaddour et al., 2023; Song et al.; Li et al., 2023a; Wang et al., 2023a; Chen et al., 2023; Zheng et al., 2023; Wang et al., 2023c). However, even the most advanced LLMs face challenges in complex multi-step mathematical reasoning problems (Lightman et al., 2023; Huang et al., 2023). To address this issue, prior research has explored different methodologies, such as pre-training (Azerbayev et al., 2023), fine-tuning (Luo et al., 2023; Yu et al., 2023b; Wang et al., 2023b), prompting (Wei et al., 2022; Fu et al., 2022), and verification (Wang et al., 2023d; Li et al., 2023b; Zhu et al., 2023; Leviathan et al., 2023). Among these techniques, verification has recently emerged as a favored method. The motivation behind verification is that relying solely on the top-1 result may not always produce reliable outcomes. A verification model can rerank candidate responses, ensuring higher accuracy and consistency in the outputs of LLMs. In addition, a good verification model can also offer invaluable feedback for further improvement of LLMs (Uesato et al., 2022; Wang et al., 2023b; Pan et al., 2023).

The verification models generally fall into the outcome reward model (ORM) (Cobbe et al., 2021; Yu et al., 2023a) and process reward model (PRM) (Li et al., 2023b; Uesato et al., 2022; Lightman et al., 2023; Ma et al., 2023). The ORM assigns a confidence score based on the entire generation sequence, whereas the PRM evaluates the reasoning path step-by-step. PRM is advantageous due to several compelling reasons. One major benefit is its ability to offer precise feedback by identifying the specific location of any errors that may arise, which is a valuable signal in reinforcement learning and automatic correction. Besides, The PRM exhibits similarities to human behavior when assessing a reasoning problem. If any steps contain an error, the final result is more likely to be incorrect, mirroring the way human judgment works. However, gathering data to train a PRM can be an arduous process. Uesato et al. (2022) and Lightman et al. (2023) utilize human annotators to provide process supervision annotations, enhancing the performance of PRM. Nevertheless, annotation by humans, particularly for intricate multi-step reasoning tasks that require advanced annotator skills, can be quite costly, which hinders the advancement and practical application of PRM.

To tackle the problem, in this paper, we propose an automatic process annotation framework. Inspired by Monte Carlo Tree Search (Kocsis & Szepesvári, 2006; Coulom, 2006; Silver et al., 2016; Świechowski et al., 2023), we define the quality of an intermediate step as its potential to deduce the correct final answer. By leveraging the correctness of the answer, we can automatically gather step-wise supervision. Specifically, given a math problem with a golden answer and a step-by-step solution, to achieve the label of a specific step, we utilize a fine-tuned LLM to decode multiple subsequent reasoning paths from this step. We further validate whether the decoded final answer matches with the golden answer. If a reasoning step can deduce more correct answers than another, it would be assigned a higher correctness score.

We use this automatic way to construct the training data for MATH-SHEPHERD, and verify our ideas on two widely used mathematical benchmarks, GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021). We explore the effectiveness of MATH-SHEPHERD in two scenarios: 1) verification: MATH-SHEPHERD is utilized for reranking multiple outputs generated by LLMs; 2) reinforcement learning: MATH-SHEPHERD is employed to reinforce LLMs with step-by-step Proximal Policy Optimization (PPO). With the verification of MATH-SHEPHERD, a series of open-source LLMs from 7B to 70B demonstrates exceptional performance. For instance, the step-by-step PPO with MATH-SHEPHERD significantly improves the accuracy of Mistral-7B (77.9%→84.1% on GSM8K and 28.6%→33.0% on MATH). The accuracy can be further enhanced to 89.1% and 43.5% on GSM8K and MATH with verification. DeepSeek 67B (DeepSeek, 2023) achieves accuracy rates of 93.3% on the GSM8K dataset and 48.1% on the MATH dataset with verification of MATH-SHEPHERD. To the best of our knowledge, these results are unprecedented for open-source models that do not rely on additional tools.

Our main contributions are as follows:

- 1) We propose a framework to automatically construct process supervision datasets without human annotations for math reasoning tasks.

- 2) We evaluate our method on both step-by-step verification and reinforcement learning scenarios. Extensive experiments on two widely used mathematical benchmarks - GSM8K and MATH, in addition to a series of LLMs ranging from 7B to 70B, demonstrate the effectiveness of our method.
- 3) We empirically analyze the key factors for training high-performing process reward models, shedding light on future directions toward improving reasoning capability with automatic step-by-step verification and supervision.

2 RELATED WORKS

Improving and eliciting mathematical reasoning abilities of LLMs. Mathematical reasoning tasks are one of the most challenging tasks for LLMs. Researchers have proposed various methods to improve or elicit the mathematical reasoning ability of LLMs, which can be broadly divided into three groups: 1) *pre-training*: The pre-training methods (OpenAI, 2023; Anil et al., 2023; Touvron et al., 2023; Azerbayev et al., 2023) pre-train LLMs on a vast of datasets that are related to math problems, such as the Proof-Pile and ArXiv (Azerbayev et al., 2023) with a simple next token prediction objective. 2) *fine-tuning*: The fine-tuning methods (Yu et al., 2023b; Luo et al., 2023; Yue et al., 2023; Wang et al., 2023b; Gou et al., 2023) can also enhance the mathematical reasoning ability of LLMs. The core of fine-tuning usually lies in constructing high-quality question-response pair datasets with a chain-of-thought reasoning process. and 3) *prompting*: The prompting methods (Wei et al., 2022; Zhang et al., 2023; Fu et al., 2022; Bi et al., 2023) aim to elicit the mathematical reasoning ability of LLMs by designing prompting strategy without updating the model parameters, which is very convenient and practical.

Mathematical reasoning verification for LLMs. Except for directly improving and eliciting the mathematical reasoning potential of LLMs, the reasoning results can be boosted via an extra verifier for selecting the best answer from multiple decoded candidates. There are two primary types of verifiers: the Outcome Reward Model (ORM) and the Process Reward Model (PRM). The ORM allocates a score to the entire solution while the PRM assigns a score to each individual step in the reasoning process. Recent findings by (Lightman et al., 2023) suggest that PRM outperforms ORM. In addition to verification, reward models can offer invaluable feedback for further training of generators (Uesato et al., 2022; Pan et al., 2023). Compared to ORM, PRM provides more detailed feedback, demonstrating greater potential to enhance generator (Wu et al., 2023). However, training a PRM requires access to expensive human-annotated datasets (Uesato et al., 2022; Lightman et al., 2023), which hinders the advancement and practical application of PRM. Therefore, in this paper, we aim to build a PRM for mathematical reasoning without human annotation, and we explore the effectiveness of the automatic PRM with both verification and reinforcement learning scenarios.

3 METHODOLOGY

In this section, we first present our task formulation to evaluate the performance of reward models (§3.1). Subsequently, we outline two typical categories of reward models, ORM and PRM (§3.2). Then, we introduce our methodology to automatically build the training dataset for PRM (§3.3), breaking the bottleneck of heavy reliance on manual annotation in existing work (Uesato et al., 2022; Lightman et al., 2023).

3.1 TASK FORMULATION

We evaluate the performance of the reward model in two scenarios:

Verification Following (Lightman et al., 2023), we consider a best-of-N selection evaluation paradigm. Specifically, given a problem p in the testing set, we sample N candidate solutions from a generator. These candidates are then scored using a reward model, and the highest-scoring solution is selected as the final answer. An enhanced reward model elevates the likelihood of selecting the solution containing the correct answer, consequently raising the success rate in solving mathematical problems for LLMs.